

ActInf Textbook Group ~ Cohort 4 ~ Meeting 22

(Chapter 10 part 2)

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all right welcome back cohort 4 this is our second discussion of chapter 10 next week will be our final week overall so we'll talk about um future textbook groups give a little time or people can fill out the forms now for feedback they can add feedback there's a lot of other um pieces that we can talk about but this will be the last recorded conversation so nice times let's go to 10 and um does anybody want to begin with any question or we'll look at what's already been written um so so Daniel I to kind of General um since I'm new to this process one of the questions that I wish I had asked you sooner was it as we uh participate and contribute to in this process um are there certain things that you would like to see that the authors would like to see um so what I'm wondering about like in terms of feedback that we can add uh through time is you know are they concerned about you know what what audiences they're trying attempting to reach um are they concerned about what principles what concepts might go over people's heads uh and make them TR away col difference prematurely um so that's one question that's General the second question I have is um and I put I added a row in theoda um I was thinking that if one of the objectives of this book is to have people who don't have the physics background walk away thinking I would like to apply this to the work that I'm doing and of course this is very selfish those S very Sly describing myself um so if if we don't want to lose people could there be a document somewhere and if it exists let me know where basically have has a list of okay in order for to understand to fully understand and I'm just going to say the free energy Principle as an example in order to fully understand the free energy principle here is a pathway of knowledge that you need to complete and it goes in a sequence from you know concept A to concept n and then uh you have the concept the equations if if the concept comes with equations and then just very succinctly described you need to know this and you need to know this and then links to videos and resources online so that basically a person can follow a self-directed learning pathway and one of the things I've done over the past week is just I decided I want to take a deep dive into the free energy

principal where did it come from who came up with it how did they use it do they use it in physics in engineering Etc and I found that it's really helpful so that was my main com word today thanks okay I'll give uh short responses to each and then anyone else can add so what what what with the authors like par and Pula and friston I mean they're happy that were enlivening and and working with the textbook um we started doing it obviously like the first day that we could and we've given them feedback on little typos and minor things but also people have engaged with them on more substantial topics and [Music] um they uh as authors and as the Publishing Company they're they're at least open to our proposals there's no questions in the textbook um and and that one textbook isn't the be all and all so it's definitely I think great to see how we can serve the authors and also the broader Niche and um in terms of like all backgrounds and the generalized learning curriculum like this textbook group is low efforts All Things Considered we don't have to prepare the materials and especially now having gone through it multiple times we um just show up and enjoy but the broader project of making knowledge spaces so that people can have self-guided and have that be like um across different languages and Technical backgrounds I mean that's the educational project that the Institute is working on so for that we just need people who are seriously committed to actually engage in that um at a slightly smaller scale Ali I believe um in the resources page this list of papers goes from from less formal to more formal but this is an infinite regress you're going to find that you keep on pulling back to to fundamental concepts which which may be good um but yes the textbook group is not the total educational Paradigm it's just one offering and we eagerly look for people who want to stand up the textbook group and see what it can be all the lwh hanging fruit like filling in the natural language descriptions of the equations and curating the discourse and adding more notes and all these things and then also the reach goals for the textbook group um like developing the software repositories that have the examples and all these other things and then even the broader educational goals that go outside of the textbook group thank youate any anyone else have a I thought on that yeah oi yeah so uh I believe being um a relatively new uh discipline and uh even a new uh I mean research Niche uh I don't think active influence has reached its full uh form yet in terms of being totally structured uh around a definite road map and so on so uh it's a bit like neuroscience at large so we know that uh unlike many other disciplines such as I don't know physics or mathematics uh which have uh a relatively more structure to uh the knowledge base they have uh in such disciplines as Neuroscience or an active influence or many other um cognitive science and uh psychological Sciences as well uh we we don't have this definite road map road maps yet uh so one way to

approach active inference is um to somehow uh determine the target topic we try to uh we try to study using active inference framework and then try to find kind of romantic or um multi Branch ways to uh approach or at least converge onto that topic so if we expect uh the learning path of active inference to be a totally linear and coherent pathway uh getting us from point A to point B and Beyond uh I I believe uh we might uh we might be disappointed at finding out that there isn't any such uh road map yet I mean uh obviously in U coming years there might be more structure to uh the learning path of active inference but uh as it's typical for many other new disciplines and research areas we're we're still exploring the forests and uh we don't have uh the cartography of the whole area yet at our disposal um thank you yeah that I make that makes sense um I think so I I understand that because fully um so I would say that in my previous conceptual work uh primarily in evolutionary biology andology um and you know the extended uh evolution of synthesis I think people would say for example I want to if I want to apply the concepts of n construction to my new to my field is there a linear time no there is you know um if I want to include everything that's under the extended evolutionary sythesis no there's no it's completely um it's quite tough actually to connect with the big areas that people have work them but um I think there might be room somewhere in in the active in Institute to to start thinking about at least like the very basic fundamental concepts so I think that if somebody sat them down and said I want you to put you know do this for the extended Evolution syes I think I could sit down and say if you don't know these things you're going to be completely lost in this little CH so there may be some foundational aspects to um in this framework that could be made accessible so to me it's all about accessibility because so why am I so concerned with this um it's because I've been looking for 20 years for a conceptual framework that I feel actually addresses aspects and phenomena that I have been looking at for this long um in what I call regenerative population health the focus in humans and the active inference framework is the first framework I have found that actually I can tell has the potential for allowing me to talk about things I could even been needing to talk about before but things that I can actually be used to and um so in my case since I my ultimately have a very fly aspect to the this which is um I work on population Health uh projects in education and hopefully in communities there's a certain urgency that I feel that I I Can't Go For example into an audience of Public Health scientists without being able to say something about why it's important to understand and that's what I'm yeah what I'm struggling with yeah we we're we're like making a thousand small tracers with people's questions here no questions no tracers a thousand questions a thousand tracers we can use curation and attention as well as language

models to synthesize what we already have written into many of the bases now there's actually two components what you ask which is the necessary and sufficient information and the sequencing sequencing in linearization is a separate question than what are the fundamentals the fundamentals are what are covered in this textbook sanj's textbook the literature those are the topics that we return to again and again now they might be challenging which then beets the question of learner specific sequencing and context specific sequencing having the situational awareness to know what you need to go into or not when you're conveying active inference for certain audience but yes we as part of a broader educational project outside of just the textbook group can use the ontology and other techniques to develop knowledge systems that that let us address these questions we're very very recently in the game and with very very few people months dedicated to the broader educational project but if it's something that people want to do there's always a niche for people who want to contribute to active inference education by doing something that's much like the textbook group like for cohort 6 or something or by diving back into the work that people have done over the years on the broader question like the fundamentals here that Brock started um I guess a year ago or the other education kind of first takes it's just a big Journey but again with our knowledge engineering project and our literature compilations and all these other things it's the the challenge that we'll address yes I there are so many I'm finding all kinds of resources as you're talking thank you appreciate any um thoughts on 10 or or any other questions one Sean you asked about the computational complexity as far yes yes I did yeah yeah as far as I know the best Oli please correct if we have not a different information but I believe that the best information was the computational complexity right here in the branching time active inference paper and model stream uh the computational model stream 6.1 branching time active inference paper here's 21 minutes in and actually the complexity of of uh active inference is like it's almost what you'd expect is it EMP hard problem or it's just uh I'm I'm not able to compare it to NP and sat and all of that but just at the within the um computational resources space I mean it's okay it's the policy Horizon the time Horizon depth times um how many policies you have times how much time times how many hidden states there are okay just like how many things do you have to keep track of broadly and then branching time active inferences using her istics to significantly reduce that complexity but they presented several complexity calculations in this paper and and with empirical run times yeah Ali I guess so branching time Act of inference the attempt was to reduce the um complexity class of the problems from uh exponential complexity to linear complexity uh so that was one way to do it

but uh a much more recent one actually it was from a few days ago uh is this recent paper by friston at all namely supervised structure learning in which uh they have proposed an alternative uh way to uh reduce the constrained mutual information of expected free energy so uh it's again another way to um reduce the complexity class of the active inference or um uh probably any other inference-based problems as well uh so yeah there are uh some attempts uh which um they explicitly address uh the problem of complexity of um re real situ real world active in problems but uh one of the main criticisms of uh or at least one of the main drawbacks of uh implementing active inference models in past years was exactly uh based on uh the exponential complexity of uh of um higher level generative models and U the ensuing um I mean uh intractability of the computation that resulted from that yeah yes um so yeah so when I look at the uh generative model and the optimization problem it seems to me that It suffers the same uh curse of dimensionality uh as we uh had in dynamic programming when you so basically your time uh explanation increase in the the number of states the more number of states whether it's hidden States or or exp States and also the number of policies um that so I wonder what kind of a trick they have being able to use to to reduce the um computational complexity from exponential uh to um to kind of polinomial or lar you know linear is a form of polinomial right so explanation is much much faster sorry is much slower than linear uh I mean near in terms of input you know so our input are um in in in in the um active inference model minimization problem the the model needs to use data uh which are the observations the uh dimensions of um of States uh the amount of policies yeah there's there's one issue I wonder if if someone can help me so the the ACT defines the policy as as the um the action they take at each time Point affordance action affordance is the action you take at each time Point policy is a sequence of actions for a given time Horizon right right so it's a sequence of action uh at uh at each time Point um so my understanding of policy is that they are actually uh they there are not only a function of time not there are sequence of time but also there are sequence of States depends on what states you're in uh you take uh different actions not not just the time but also uh the hiden hidden States um so I wonder if they're are the same um if what are the same you know so so uh action as a sequence of Time Versus action as a sequence of states and time and know like ideally whether um you take the action at 2: p.m. or 7: p.m. or 11:00 p.m. like the number of the time policy selection should be invariant unless the time is the determinant factor for which policy to take so the time number itself shouldn't matter and then as to whether a given observation or hidden state has a bearing on policy selection that's the question there's no reason to address that question in the abstract

because obviously we can imagine situation that do and don't have that um and also while we're in this space of computational accelerations like in the in the most recent message passing live stream now there's direct policy inference without expected free energy so I mean that's why we just hold on to our hats and stay in the game because it may not be required to iterate overall policies with an compare the expected free energies you know du research on EF if you read the paper 56 and watch the discussions with Magnus we heavily discuss this direct policy inference technique so sometimes the kind of all by all by all is like the pedagogical or just the first sketch but don't wait on the sideline because you're worried about a computational Slowdown that may or may not happen either jump in and find out about that slowdown or jump in and then you'll find out that there's no slowdown or that by the time that you get to actually building that model in three months or six months that the computational environment has evolved but the general yeah so I I'm actually I'm I'm um supervised my student to build the um build the active influence model and when uh when I so so you have to you have to Define uh the policies um what is Define State you define the afford and then the time Horizon what is the for this those are the actions that can be taken in a moment up down left right four affordances and then we're going to do you know two time steps and then there's 16 policies the policies are just the enumeration of all the possible sequences of affordances of that length so policy construction is Trivial policy evaluation traditionally has been done with the efe but now there's an even more direct way to do it doesn't mean that this is the best way to do it in every single person's situation for every single problem just that there is a technique now where you don't need to do expected free energy based comparison you can do generalized free energy based direct policy inference it just is that way yeah so I'm repeating myself uh sorry just a one last time hope somebody can help me so policy is defined as function of time right so in the sense that you see it's a sequence of action that's a policy defined in the textbook um sequence of action you know at time one what you do at time two what you do left or right those actions but in general in my area in my uh dynamic programming area the parties are defined as a function of State not not only time but also State uh in our example for example if whether depends on what whether you're hungry or not hungry if one of the state of your senses uh data is that whether you're hungry or not then you know your PES actually depends on whether you're going to go for food or or just just take a bite of some food depends on your state you're the hungry or not hungry State you know not necessarily depends on time well yes in different areas of control terms like policy are used slightly differently but broadly we're talking about the same thing which is a policy is a

sequence of actions and you can construct the model to be borne upon by hunger or not so whether or not a given bearing on a given policy selection that's up to the modeler that's not going to be addressed by active inference textbook group generalities but yeah if you make a model where hunger Bears upon the choice to eat an apple then you have your answer if you do it where it doesn't you also have your answer okay but there isn't an in principle answer to a situational design so so what you're see that as a as a researcher I can Define the policy in different way not not uh not just the sequence of action uh over time by a sequence of action over over States over um hidden States for example as part of constructing the entire generative model so a b CDE e and so on you're right you decide not just the affordances and the time Horizon which Define the policy space but you also so establish the sparsity and the connections of what else is included so if it's just a thermometer and then the thermometer is unrelated to um you know a slot machine player that's what you have but if the slot machine's conditioned on temperature that's what you have the slot machine's conditioned Upon A Taste sensor that's what you have so it's like a toolbox for us to take those steps and ask what if but we can't return to the framework and ask about whether certain factors influence each other or not again because we can just quickly think about examples where they wouldn't wouldn't that's what chapter six is about is defining the whole generative model not just we don't just hardcode policies yeah thanks I I need to think a more how to how to U generalize the the policy you know to extend the model in chapter six uh to more General uh policy definition okay thank you anyway for uh clar oh yes this um let's look at a few of the questions this is a great great starting one section 10.4 there's a song lyric and then this person has written another verse from that song that's a little bit less juvenile violence in theme okay anyone W to yeah yeah go ahead yeah I just wanted to I I need some clarification and I think it I don't know if it's relevant to um L's question um but there are states hidden states that we attempt to infer from um our models that's why understanding then there's perception and then there's action and one of the things that um the themes in the book is that action and perception uh if I understood this correctly are seen as one and the same thing could you elaborate on that a lot of great questions I you said that that through modeling like we we find or we discover hidden States a slightly more because we can't view them we can't see them we have to infer them right we have a microcosm of the active agent which is that the active agent is only getting its sensorium the inbound and its its actions on the outbound and it is proposing latent unobserved hidden causes of sensory phenomena to bound its surprise so um in in this kind when we write this kind of a model like figure 4.3 we put the hidden State down temperature

in the room we put the observation down thermometer and then the agent does the inference on the hidden state so but but we constructed the hidden state space to be fixed or structure learning is like when the agent is going to be like rearranging combinations of hidden states and and learning the structure of the world for itself um action and perception are part of a unified unity is plural at minimum too perception is not action simply now if you look across a blanket one agent's action becomes the other's perception so in that sense action and perception are just kind of like whose perspective are you taking from the blanket like you know screaming into the environment is the action from the yellor but then it's the perception from the environment so that's one sense but um action and perception are unified in act of inference not by eradicating their differences but by understanding their complete articulation which is the two directions of traffic across the blanket so it's not that you know listening is the same as kicking your leg it's not that perception is action it's that they're part of a unified cognitive framework that has a unified imperative with free energy minimization unified handling of variables like this and so on okay so having said that if you if having said that for one individual agent that when we move into really complex systems like the social insects you work who for a complicated system like an ants nest the system is the ants nest can you tell me just plain English what's this is what are the actions examples of actions and examples of perceptions in that system um sure well although the ant colony system is like the evolutionary replicator or or the Conan organism in the active inference model we made an agent-based model with an estimate and we used exactly the figure 4.3 type generative model right perception was pherone and action was movement so broadly the the the selection of the perception and action of the agent is similar to any other agent-based modeling technique if you model the ant colony as a single organism that's a totally cool and valid model there's always a plurality of models available so modeling a single colony as the agent maybe if you wanted to model like a field site with just several thousand colonies and you didn't want to go into the estimate level of detail or maybe the estimate level is too coarse grains and maybe you need to do an organ specific model these are maps not territories so it's whatever you need to do to take the next step on your path not about there only being like one ants model right so then so then in in so as systems so systems to me are just I can't think of any system that at least in the in a phenomena that I'm familiar with where there's anything any individual unit because even if we go down to a cell a cell is a system you know or if we go down to a microb the microb is a system and it's they're very complex system so so it just seems to me that there are systems and nested systems within systems and interacting systems and depend all kinds of

systems interacting with each other Etc so um so I know there are different ways of thinking about complexity uh I think about it right now in a very qualitative way but it seems to me that like for example in the ants nest Nest you have all kinds of systems going you have the system of the you know maybe the um the queen and everything that's associated with her reproduction then they're the systems of all the workers do you know U keeping the dro clean and so on and so forth so there are so many ways that you can take these Concepts and and I identify markco Mark of blankets that are relevant to different agents within systems and that can be then blown up to a whole system in in my understanding this corly yeah yeah I mean at the most zoomed out that's also true of linear modeling you could say well we could do a linear aggression of the Queen's femur versus the number of seeds per minute per day squared you know there's no limit to the amount of linear aggressions um fmri analysis so this just science and instrumentalism as a technique there's never any ending to how many models can be this this is not active specific now you said there's nested interacting blankets or nexted an interacting um agents also very true we have enclosures of systems and laterally interacting Collective Behavior but those are expressions about the territory you're talking about how the colony really is you could choose to model it in different ways an active inference gives us a qualif IED way to stop that infinite regress by saying yeah there there are organs within the nestmate they do have a midgut and a brain and all this we're modeling this level and then that's the perfect stopping point for then a future study to pick up there and go one level deeper into the blanket so it's not that one study must make all possible models and Ma of all detail it's actually leading to a more principled way of collaborating across these different systems that's the goal of the infer ants and and analogously for other systems I mean why do we have so many um Square One models of ant foraging but there's never like sustained development of multiple Labs with multiple species looking at different foraging environments because it's easy to make that custom model of that one ant in that one setting but then to generalize it to multiple ant species or multiple settings let alone tasks other than foraging it's very challenging so then the literature is basically littered with kind of toy models or intuition pumps but there's never a strong substrate to build a real extensible model like a Linux kernel for agent based simulations and and those have been tried but then those systems that try to generalize agent based simulations themselves get mired down in technical details and and frictions of accessibility but you've definitely identified the key aspects of what we can explore and that's why chapter six is provided chapter 6 doesn't say here's how nature is organized chapter 6 says here's how you build that generative model and then how does

chapter 10 end build the model and get surprised it doesn't end with great now you've established how nature is got it so so my next I go with my stupid question my next stupid question but I have to ask it yeah because it seems like throughout the book there's like this this in some places is very clear and I've seen it in other contexts where it's stated clearly that what what the the goal is to to take this understanding this deep structure based on the concepts and the architecture of the ecosystem of active inference whatever it looks like you know for whomever but there's kind of a consensus around certain ideas that hold this conceptual framework together but the idea is that to be able to take that and then go to into artificial intelligence and machine learning and say this stuff that we see in living organisms this deep structure we can translate it into a completely artificial system and recreate uh actions um by translating this deep structure into something that works in a machine um and then if that's the case then I guess for me as a social scientist is the Deep inference here that there is really no boundary between living and non- living systems from a active influence perspective there's a lot to say on that um one could debate whether the construction of a synthetic anything synthetic chemistry synthetic drawing a synthetic active inference agent does constructing a synthetic anything legitimate any Claim about the territory just making an arts and crafts project doesn't change the nature of the trees and the sunset so that's one point second point would be yeah instrumentally if not further we have a strong continuity of diverse systems inert systems sleeping systems active systems and so on so from a instrumental descriptive perspective we have an integrated modeling approach across different kinds of systems again this doesn't mean life and non-life are the same exactly it means that there's a pallet for people to say this is what I mean by life and I made models that are on either side of the line and this one's in the gray Zone and someone can say well my Line's over here well that's just the fact of the matter then that those people have those lines yes okay thank you appreciate it yeah these are these are let's see if we can look at just just so that we can okay I'll start on the bottom okay yes thank you for this one addressed earlier yes join everyone is encouraged to join our broader educational efforts and to enrich the textbook group where we actually will do these things um which sections discuss is basing computation the most effective problem solving system for minimizing free energy not sure if in the book per se they might discuss basian optimality um other people I know elsewhere take more like kind of first principles here's why things should be or ought to be basian but suffice to say the framework doesn't rest upon basian computation being the most effective this is just a framework that we can use to describe any system now what's the most effective to actually do

it that may or may not be doing Bayesian statistics but the analytical formalism will never fail us as observers but that doesn't mean it's the best way to build it from the inside it'd be helpful to provide definitions for thermodynamics free energy and of ACM variational free energy throughout the book free energy is used interchangeably with variational free energy in this book there are probably un less explicitly mentioned they're only talking about variational free energy they're never talking about Gibbs unless they specifically called it out and I don't think they do they're only concerned with informational free energy not thermal sections on the the term of free yeah yeah definitely there's always ways to connect it back and um let's go to the strongest Alex Kefir psychophysical identity and free energy live stream 22 this is the strongest statement or one of the strongest statements made that that the thermodynamic free energy of a system directly encodes variational free energy so we're walking that pedagogical line you know they're different concepts you could link them how you want and then Kefir just runs in and says no there the same or there are systems for which the thermodynamic free energy flows do entail variational free energy flows okay definition of brain definitions for mind brain Behavior yeah I mean there's there's the author's definition and then there's our ontology and we meet every week to update the definitions and to translate them and to use them more coherently so people especially are interested in definitional questions Let's Do It um brains and centralized nervous systems not exactly sure what you mean by the brain is a recent development I mean there going back to the last common ancestor of vertebrates and invertebrates there was a centralized nervous system and it changed in in structure and function even in the last tens of thousands of years uh uh in that question it's kind of along the same lines of um the what's living and what isn't um in the so I guess you know I what you're right I mean it's it's about you as the Observer and researcher you decide where you you know what what how you're going to Define things what your affordances are what is the system that you're working with but I when I was first reading through the book I was just thinking are they is this only referring to systems where you have to have a nervous system and then what I was talking about is that the the human brain in the in terms of the neurological variation that's out there in nature it just has um you know um very different structure very complex structures um anyway so it's along the line so you know do you have to have this neurological kind of sensorium in order for to use active in models and I think you guys that that good question it absolutely does not require centralized nervous system or nervous cells it could be a pendulum it could be in a ner Rock on the ground it doesn't need to be any kind of biological life form or any kind of anatomy thank you yeah that's really great but again but

most of the work has arisen from Carl Friston and colleagues doing human neuroimaging again every single time in chapter five that's what we emphasize chapter 4 generative model in SS chapter 5 they go to the case and the setting where the most work empirically has been done which is human Neuroscience but there's so so many other ways to go thank you okay not is this a question or or what about this can't really see it's too small opportunity to expand on Niche construction I I agree with this okay yeah I mean the the um as everybody you know that sequencing question we talked about earlier like the textbook is just one sequence and so some people are going to say connect it more to what has empirically been done and I want to understand how brains work I want to have that first person driver seat perspective I want the stories to be about me picking up the water me getting out of the chair me listening to music other people are going to say isn't it about bigger than humans or different than humans shouldn't we use only the technical mathematical aren't um appeals to personal experience possibly even misleading everyone's right that's why we need a bigger knowledge system that can generate those sequenced paths so if people want the human 101 no math in this language they can have that path and if somebody wants a different language different technical level no human examples that is also going to be consistent um and so is is um so Chris Field's um course on information um physics as processing is that you know in this context um trying to connect thoughts here um is that um a way of saying or his perspective is we don't have to make the work so complicated at the quantum from a quantum physics perspective we can actually think about active infl a way is in my and trying to get at the the deeper structure of information process yeah this this textbook was written probably largely between 2018 and 2022 maybe even 2021 because it was published in March 2022 and it was centered around Princeton's 2019 particular physics um on graph developed a lot the quantum free energy principle work with Fields glazebrook Levan Marciano at all only was in 2022 live stream 40 so this textbook does not include any Quantum informatics Chris fields and colleagues with their backgrounds in the quantum information Sciences made that connection between free energy principle and quantum so that connection is basically unaddressed in this textbook only only in the most minimal possible way hinting at the Bayesian mechanics so this textbook is um I mean those who love the history and philosophy of science again let us reconvene to write these histories but this book is in some ways like a coda on the 2010s of active inference because already in the 2020s we're heading off in some new directions and so it's kind of like an inflection point that Consolidated a huge amount of the past work but now with message passing and Julia on this side and Quantum fvp on that side and far more empirical

applications happening all around it's a different environment for doing um active inference learning and application and for writing a textbook as we could ask sanjiv about so this textbook is not everything and again just from a chapter 10 perspective in only about 200 Pages 200 very short Pages what they've done is incredible and that's why we can always return to it and and um yeah even though it's only been 18 months since it's being released but this is the discussion we've had that does not conclude cohort 4 I hope that people return um next week to to discuss more and and um yeah update things develop things add projects add questions develop the answers thank you everybody oh thank you appreciate it see you next time see you next time thank you Daniel bye bye byebye